

CH 2 - GDP and its Components

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2.1 GDP Definition

Definition – GDP

GDP is the **market value** of all **final goods and services** produced within a country during a given period

$$GDP = \sum_i P_i Q_i$$

- It measures a **flow**: production over a month, quarter, or year
- It measures **economic activity**, not the **stock** of national wealth
- It is measured to support comparisons across time and across economies.

Market Value: output is recorded using market prices.

Final Output: Intermediate goods are not counted

Domestic production: production counts where it occurs, regardless of ownership

Included in GDP	Excluded from GDP
Paid childcare services	Unpaid childcare and household work
Newly produced market goods and services	Volunteer services
Government services valued at cost	The resale value of used goods
Imputed rent for owner-occupied homes	Many benefits from free digital services

Note: Unreported market activity is conceptually included in GDP, but it is difficult to observe and may be underestimated

Housing

Included in GDP	Excluded from GDP
New home construction and major renovations	Purchase price of an existing home
Real-estate, legal, and inspection services associated with a sale	A capital gain, or increase in the owner's housing wealth
Rent and the imputed rental value of owner-occupied homes	

Own house → Imputed rent → Net amount → GDP

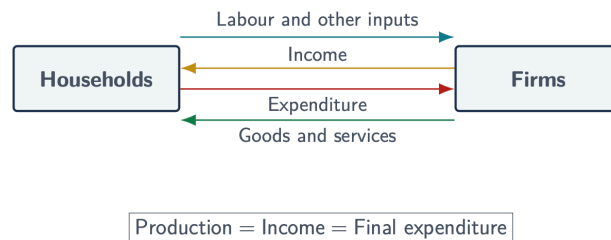
Note: GDP records current housing production and services, not changes in the value of existing property

2.2 Measurement of GDP

Three methods of measuring GDP:

1. **Production method(value-added)**
2. **Income method**
3. **Expenditure method**

The Circular Flow Connects the Three Approaches



Info – Production Method

Value-added Method

Gross value added – value of output – cost of intermediate inputs

- Intermediate inputs are goods and services purchased from other firms and used to produce current output
- Adding the value created at each stage of domestic production measures total value added without counting intermediate inputs more than once.

From value added to GDP: StatCan adds taxes on products and subtracts subsidies on products when calculating the published GDP total

Examples:

1. Suppose we have a simple economy:

- Farmer sells wheat to miller for 80
- Miller sells flour to baker for 140
- Baker sells bread to consumer for 220

We have:

$$\begin{aligned}\text{Farmer} &: 80 - 0 = 80 \\ \text{Miller} &: 140 - 80 = 60 \\ \text{Baker} &: 220 - 140 = 80\end{aligned}$$

$$\text{GDP} = 80 + 60 + 80 = 220$$

It is not a coincidence the GDP is same as the price of the bread.

2. Suppose we have another economy measured in thousands of dollars

- Forestry company: sells logs to a CAN sawmill for 300 and uses imported fuel worth 50
- Sawmill: buys logs and imported processing materials worth 80. Sells lumber worth 600 to a CAN furniture producer and exports another 100
- Furniture producer: buys lumber and imported components worth 150. Sells furniture worth 1100 and adds finished worth 100 to its inventories.

We have

$$\text{Forestry company} : 300 - 50 = 250$$

$$\text{Sawmill} : 600 + 100 - 380 = 320$$

$$\text{Furniture producer} : 1200 - 600 - 150 = 450$$

$$\text{Total GDP} = 1020$$

Info – Income Method

- **Compensation of Employees:** wages, salaries, and employer's social contributions
- **Gross operating surplus:** income generated mainly by incorporated business, before deducting depreciation
- **Gross mixed income:** income of unincorporated business, combining returns to the owner's labour and capital
- **Taxes less subsidies on productions and products:** taxes associated with production and sales, net of subsidies

The formula:

$$\text{Net domestic product} = \text{GDP} - \text{depreciation}$$

Note: The biggest component of the Income method is Compensation of Employees, then it is Gross operating surplus

Info – Expenditure method

C : Household consumption

I : Business fixed, residential, and inventory investment

G : Government purchases of goods and services

$X - M = NX$: Exports minus imports or Net exports

Formula:

$$Y = C + I + G + (X - M) = C + I + G + NX$$